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### Control method for sheet manufacturing apparatus

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#### Abstract

A control method for a sheet manufacturing apparatus includes a stop signal input step of inputting, to the sheet manufacturing apparatus manufacturing a sheet, a stop signal stopping manufacturing of the sheet and a drying step of drying a predetermined area of the sheet manufacturing apparatus by controlling a blower of the sheet manufacturing apparatus after the stop signal input step.

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## Background/Summary

(1) The present application is based on, and claims priority from JP Application Serial Number 2021-175295, filed Oct. 27, 2021, the disclosure of which is hereby incorporated by reference herein in its entirety.

### BACKGROUND

#### 1. Technical Field

(2) The present disclosure relates to a control method for a sheet manufacturing apparatus and a sheet manufacturing apparatus.

#### 2. Related Art

(3) A sheet manufacturing apparatus for manufacturing sheets is known. For example, JP-A-2020-143398 describes a paper processing apparatus that cleans devices by using water.

(4) However, in the paper processing apparatus using water as described above, rust may form or various biological contaminants may proliferate due to humidity in the apparatus after the manufacturing of sheets is stopped.

## SUMMARY

(5) A control method for a sheet manufacturing apparatus according to an aspect of the present disclosure includes: a stop signal input step of inputting, to the sheet manufacturing apparatus manufacturing a sheet, a stop signal stopping manufacturing of the sheet; and a drying step of drying a predetermined area of the sheet manufacturing apparatus by controlling a blower of the sheet manufacturing apparatus after the stop signal input step.

(6) A sheet manufacturing apparatus according to an aspect of the present disclosure includes: a manufacturing section manufacturing a sheet; and a control section controlling the manufacturing section, in which the control section dries the predetermined area of the manufacturing section by controlling the blower of the manufacturing section when receiving a stop signal stopping operation of the manufacturing section.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram schematically illustrating a sheet manufacturing apparatus according to a first embodiment.

(2) FIG. 2 is a flowchart for describing a control method for the sheet manufacturing apparatus according to the first embodiment.

(3) FIG. 3 is a diagram schematically illustrating the sheet manufacturing apparatus according to the first embodiment.

(4) FIG. 4 is a diagram schematically illustrating a sheet manufacturing apparatus according to a second embodiment.

(5) FIG. 5 is a flowchart for describing a control method for the sheet manufacturing apparatus according to the second embodiment.

(6) FIG. 6 is a diagram schematically illustrating the sheet manufacturing apparatus according to the second embodiment.

(7) FIG. 7 is a diagram schematically illustrating the sheet manufacturing apparatus according to the second embodiment.

(8) FIG. 8 is a diagram schematically illustrating a sheet manufacturing apparatus according to a third embodiment.

(9) FIG. 9 is a flowchart for describing a control method for the sheet manufacturing apparatus according to the third embodiment.

(10) FIG. 10 is a diagram schematically illustrating a sheet manufacturing apparatus according to a fourth embodiment.

### DESCRIPTION OF EXEMPLARY EMBODIMENTS

(11) Preferred embodiments of the present disclosure will be described in detail below with reference to the drawings. It should be noted that the embodiments described below do not unreasonably limit the content of the present disclosure described in the claims. In addition, not all of the components described below are essential structural requirements for the present disclosure.

#### 1. First Embodiment

(12) 1.1 Sheet Manufacturing Apparatus

(13) 1.1.1 Overall Structure

(14) First, a sheet manufacturing apparatus according to a first embodiment will be described with

reference to the drawings. FIG. 1 is a diagram schematically illustrating the sheet manufacturing apparatus **100** according to the first embodiment.

(15) As illustrated in FIG. 1, the sheet manufacturing apparatus **100** includes a manufacturing section **110** and a control section **170**. The manufacturing section **110** manufactures a sheet S. The manufacturing section **110** includes, for example, a supply section **10**, a coarse crushing section **12**, a defibering section **20**, a sorting section **30**, a first web forming section **40**, a first moisture adding section **44**, a rotating body **47**, a mixing section **50**, an accumulation section **60**, a second web forming section **70**, a second moisture adding section **74**, a transport mechanism **77**, a sheet forming section **80**, a cutting section **90**, an ejection reception section **94**, and a humidifier **120**.

(16) The supply section **10** supplies a raw material to the coarse crushing section **12**. The supply section **10** is, for example, an automatic feeding section for continuously feeding the raw material to the coarse crushing section **12**. The raw material supplied by the supply section **10** includes, for example, fibers such as used paper and pulp sheets. In the illustrated example, the raw material is transported to the coarse crushing section **12** via a transport roller **11**.

(17) The coarse crushing section **12** cuts the raw material supplied by the supply section **10** into strips in a gas such as air. The strips are, for example, several centimeters long. In the illustrated example, the coarse crushing section **12** has a coarse crushing blade **14** and cuts the fed raw material with the coarse crushing blade **14**. The coarse crushing section **12** is, for example, a shredder. The raw material cut by the coarse crushing section **12** is received by a hopper **1** and then transported to the defibering section **20** through a pipe **2**. The pipe **2** is provided with a blower **102**. The blower **102** generates an air flow that transports the cut raw material to the defibering section **20**.

(18) The defibering section **20** defibers the raw material cut by the coarse crushing section **12**. Here, “defibering” denotes unraveling of the raw material including a plurality of bound fibers into individual fibers. The defibering section **20** further has a function of separating, from fibers, substances such as resin particles, ink, toner, and bleed inhibitors which have adhered to the raw material.

(19) Substances having passed through the defibering section **20** are called defibered substances. Defibered substances may include resin particles, coloring agents such as ink and toner, which have separated from fibers when the fibers are unraveled, and additives such as bleed inhibitors and paper strength enhancers, in addition to unraveled defibered substance fibers. The unraveled defibered substance is shaped like a string. The unraveled defibered substance may be present in an independent state in which it is not entangled with other unraveled fibers or may be present in a cluster state in which it is entangled with other unraveled defibered substances to form a cluster.

(20) The defibering section **20** performs dry-type defibering. Here, “dry type” denotes processing such as defibering or accumulation performed in a gas such as air, rather than in a liquid. For example, an impeller mill is used as the defibering section **20**. The defibering section **20** has a function of generating an air flow that sucks the raw material and discharges the defibered substance. This enables the defibering section **20** to suck the raw material through the inlet **22** together with an air flow generated by the defibering section **20**, defiber the raw material, and transport the defibered substance to the outlet **24**. The defibered substance having passed through the defibering section **20** is transported to the sorting section **30** through a pipe **3**.

(21) It should be noted that the air flow for transporting the defibered substance from the defibering section **20** to the sorting section **30** may be the air flow generated by the defibering section **20** or the air flow generated by a blower **103** to be provided in the pipe **3**.

(22) The sorting section **30** introduces, through an inlet **32**, the defibered substance defibered by the defibering section **20** and sorts the defibered substance according to the fiber length. The sorting section **30** has, for example, a drum section **31** and a housing section **33** that accommodates the drum section **31**. For example, a sieve is used as the drum section **31**. The drum section **31** has a mesh. The drum section **31** can separate the introduced defibered substance into a first sorted

substance that passes through the mesh and a second sorted substance that does not pass through the mesh. The first sorted substance is a fiber or particle that is smaller than the size of the openings of the mesh. The second selection is a fiber, an undefibered piece, or a cluster that is larger than the size of the openings of the mesh. The first sorted substance is transferred to the accumulation section **60** through a pipe **4** and a pipe **5**. The second sorted substance is returned to the defibering section **20** from an outlet **34** through a pipe **6**. Specifically, the drum section **31** is a cylindrical sieve that is rotationally driven by a motor. For example, a wire mesh, an expanded metal mesh obtained by stretching a metal plate having openings, or a punching metal mesh having holes in the metal plate formed by a press machine or the like is used as the mesh of the drum section **31**.

(23) The first web forming section **40** transports the first sorted substance having passed through the sorting section **30** to the pipe **5**. The first web forming section **40** includes, for example, a mesh belt **41**, a stretching roller **42**, and a suction mechanism **43**.

(24) The first sorted substance having passed through the openings of the sorting section **30** is accumulated on the mesh belt **41**. The mesh belt **41** is stretched by the stretching roller **42** and has a structure that does not easily pass the passed-through substance but easily passes air. The mesh belt **41** moves when the stretching roller **42** rotates. The first sorted substance having passed through the sorting section **30** continuously accumulates on the mesh belt **41** while the mesh belt **41** continuously moves, thereby forming a web **V** on the mesh belt **41**.

(25) The suction mechanism **43** is provided below the mesh belt **41**. The suction mechanism **43** generates a downward air flow. The first sorted substance dispersed in air by the sorting section **30** can be sucked onto the mesh belt **41** by the suction mechanism **43**. This can increase the speed of discharge from the sorting section **30**. Furthermore, the suction mechanism **43** can form a downflow in the drop path of the first sorted substance and prevent the defibered substances from being entangled with each other during the drop.

(26) As described above, passing through the sorting section **30** and the first web forming section **40** causes the web **V** to contain a large amount of air and become soft and swollen.

(27) The first moisture adding section **44** adds moisture to the web **V** on the mesh belt **41**. The first moisture adding section **44** includes, for example, a mist humidifier **45** and a suction mechanism **46**. The mist humidifier **45** forms a mist from water and adds moisture to the web **V** on the mesh belt **41**. The mist humidifier **45** is, for example, an ultrasonic humidifier. The suction mechanism **46** is provided below the mesh belt **41**. The suction mechanism **46** generates a downward air flow. The suction mechanism **46** can efficiently add the moisture from the mist humidifier **45** to the web **V**. The moisture having passed through the web **V** is discharged to the outside through a pipe **7**. The pipe **7** is provided with a blower **107**. The blower **107** generates an air flow for discharging the moisture to the outside. The web **V** to which the moisture has been added is fed to the pipe **5** and transported to the accumulation section **60**.

(28) The rotating body **47** is provided in the pipe **5**. The rotating body **47** cuts the web **V**. In the illustrated example, the rotating body **47** has a base portion **48** and projection portions **49** projecting from the base portion **48**. The protrusion **49** has, for example, a plate-like shape. In the illustrated example, four protrusions **49** are provided at equal intervals. As the base portion **48** rotates, the projection portions **49** rotate about the base portion **48**. By cutting the web **V** via the rotating body **47**, for example, fluctuations in the amount of the defibered substance per unit time supplied to the accumulation section **60** can be reduced.

(29) The rotating body **47** is provided at a position at which the projection portions **49** can make contact with the web **V** but do not make contact with the mesh belt **41** on which the web **V** is accumulated. This can prevent the mesh belt **41** from being worn by the projection portions **49**. The shortest distance between the projection portions **49** and the mesh belt **41** is, for example, not less than 0.05 mm and not more than 0.5 mm.

(30) The mixing section **50** mixes the first sorted substance having passed through the sorting section **30** with an additive. The mixing section **50** includes, for example, an additive supply

section **52** that supplies the additive, a pipe **54** through which the first sorted substance and the additive are transported, and a blower **56**. The pipe **54** communicates with the pipe **5**.

(31) In the mixing section **50**, an air flow is generated by the blower **56**, and the first sorted substance and the additive are transported while being mixed with each other in the pipe **54**. It should be noted that the mechanism for mixing the first sorted substance with the additive is not particularly limited and may be a device for performing agitation by using blades rotating at high speed or a device in which a container is rotated, as in a V-type mixer.

(32) A screw feeder as illustrated in FIG. **1** or a disc feeder, which is not illustrated, is used as the additive supply section **52**. The additive supplied from the additive supply section **52** contains a resin for binding a plurality of fibers to each other. At the time the resin is supplied, the plurality of fibers are not bound to each other. The resin melts when passing through the sheet forming section **80** and binds the plurality of fibers to each other.

(33) The resin supplied from the additive supply section **52** is a thermoplastic resin or a thermosetting resin, such as AS (acrylonitrile styrene) resin, ABS (acrylonitrile butadiene styrene) resin, polypropylene, polyethylene, polyvinyl chloride, polystyrene, acrylate resin, polyester, polyethylene terephthalate, polyphenylene ether, polybutylene terephthalate, nylon, polyamide, polycarbonate, polyacetal, polyphenylene sulfide, polyether ether ketone, or the like. These resins may be used alone or in mixture as appropriate. The additive supplied from the additive supply section **52** may be in the form of fibers or powder.

(34) It should be noted that the additive supplied from the additive supply section **52** may include colorants for coloring fibers, aggregation inhibitors for suppressing the aggregation of fibers and the aggregation of resin, and flame retardants to suppress the fibers and the like from burning according to the sheet **S** to be manufactured, in addition to resins for binding fibers to each other. The mixture having passed through the mixing section **50** is transported to the accumulation section **60** through the pipe **54**.

(35) The accumulation section **60** introduces the mixture having passed through the mixing section **50** through the inlet **62**, unravels the entangled defibered substance, and drops the unraveled defibered substance while dispersing the defibered substance in air. The accumulation section **60** accumulates the defibered substance in a dry manner to form a web **W**. Furthermore, when the resin of the additive supplied from the additive supply section **52** is fibrous, the accumulation section **60** unravels the entangled resin. Accordingly, the accumulation section **60** can uniformly accumulate the mixture of the defibered substance and the additive on the second web forming section **70**.

(36) The accumulation section **60** includes, for example, a drum section **61** and a housing section **63** that accommodates the drum section **61**. A rotating cylindrical sieve is used as the drum section **61**. The drum section **61** has a mesh through which the fibers or particles that are included in the mixture having passed through the mixing section **50** and that are smaller than the size of the openings of the mesh are dropped. The structure of the drum section **61** is the same as, for example, the structure of the drum section **31**.

(37) It should be noted that the sieve of the drum section **61** does not need to have a function of sorting a specific object. That is, the sieve used as the drum section **61** is a component with a mesh, and the drum section **61** may drop the entire mixture introduced to the drum section **61**.

(38) The passed-through substance having passed through the accumulation section **60** is accumulated on the second web forming section **70** to form the web **W**. The second web forming section **70** includes, for example, a mesh belt **71**, a stretching roller **72**, and a suction mechanism **73**.

(39) The passed-through substance having passed through the opening of the accumulation section **60** is accumulated on the mesh belt **71**. The mesh belt **71** is stretched by the stretching roller **72** and has a structure that does not easily pass the passed-through substance but easily passes air. The mesh belt **71** moves when the stretching roller **72** rotates. The passed-through substance having passed through the accumulation section **60** continuously accumulates on the mesh belt **71** while

the mesh belt **71** continuously moves, thereby forming the web W on the mesh belt **71**.

(40) The suction mechanism **73** is provided below the mesh belt **71**. The suction mechanism **73** generates a downward air flow. The mixture dispersed in air by the accumulation section **60** can be sucked onto the mesh belt **71** by the suction mechanism **73**. This can increase the speed of discharge from the accumulation section **60**. Furthermore, the suction mechanism **73** can form a downflow in the drop path of the mixture and prevent the defibered substance and the additive from being entangled with each other during the drop.

(41) As described above, passing through the accumulation section **60** and the second web forming section **70** causes the web W to contain a large amount of air and become soft and swollen.

(42) It should be noted that the passed-through substance having passed through the opening of the accumulation section **60** and then passed through the openings of the mesh belt **71** is returned to the pipe **2** through a pipe **8**. The pipe **8** is provided with a blower **108**. The blower **108** generates an air flow for returning the passed-through substance to the pipe **2**.

(43) The second moisture adding section **74** adds moisture to the web W on the mesh belt **71**. The second moisture adding section **74** includes, for example, a mist humidifier **75** and a suction mechanism **76**. The mist humidifier **75** forms a mist from water and adds moisture to the web W on the mesh belt **71**. The mist humidifier **75** is, for example, an ultrasonic humidifier. The suction mechanism **76** is provided below the mesh belt **71**. The suction mechanism **76** generates a downward air flow. The suction mechanism **76** can efficiently add the moisture from the mist humidifier **75** to the web W. The moisture having passed through the web W is discharged to the outside through a pipe **9**. The pipe **9** is provided with a blower **109**. The blower **109** generates an air flow for discharging the moisture to the outside.

(44) The transport mechanism **77** transports the web W to which moisture has been added from the second web forming section **70** to the sheet forming section **80**. The transport mechanism **77** includes, for example, a mesh belt **77a**, a stretching roller **77b**, and a suction mechanism **77c**.

(45) The mesh belt **77a** makes contact with the upper surface of the web W. The mesh belt **77a** is stretched by the stretching roller **77b**. The mesh belt **77a** moves when the stretching roller **77b** rotates.

(46) The suction mechanism **77c** is provided above the mesh belt **77a**. The suction mechanism **77c** generates an upward air flow. The suction mechanism **77c** causes the web W to be separated from the mesh belt **71**, make contact with the mesh belt **77a**, and be transported to the sheet forming section **80** by the mesh belt **77a**. A pipe **13** is coupled to the suction mechanism **77c**. The pipe **13** is provided with a blower **113** that generates an upward air flow.

(47) The sheet forming section **80** forms the sheet S by pressurizing and heating the web W accumulated on the mesh belt **71**. The sheet forming section **80** can bind a plurality of fibers in the mixture to each other via the additive by heating the mixture mixed in the web W.

(48) The sheet forming section **80** includes, for example, a pressurizing section **82** that pressurizes the web W and a heating section **84** that heats the web W pressurized by the pressurizing section **82**.

(49) The pressurizing section **82** has a pair of calender rollers **83**. The pair of calender rollers **83** applies pressure to the web W. The pressure causes the web W to be reduced in thickness and increased in bulk density.

(50) The heating section **84** has a pair of heating rollers **85**. The pair of heating rollers **85** heats the web W. The sheet S can be formed while the web W is continuously transported when the heating section **84** is configured as the heating rollers **85**, compared with the case in which the heating section **84** is configured as a plate-like press device. The calender rollers **83** and the heating rollers **85** are disposed such that the rotary axes thereof are parallel to each other. The calender roller **83** can apply, to the web W, a pressure higher than the pressure applied to the web W by the heating roller **85**. It should be noted that the number of calender rollers **83** and the number of heating rollers **85** are not particularly limited.

(51) The cutting section **90** cuts the sheet S formed by the sheet forming section **80**. In the illustrated example, the cutting section **90** cuts the sheet S in a direction intersecting the transport direction of the sheet S. The cutting section **90** forms a single sheet S with a predetermined size. The single sheet S is ejected to the ejection reception section **94** via the plurality of transport rollers **92**. It should be noted that the number of transport rollers **92** is not particularly limited.

#### (52) 1.1.2 Humidifier and the Like

(53) The manufacturing section **110** of the sheet manufacturing apparatus **100** includes, for example, the humidifier **120**. Furthermore, the manufacturing section **110** of the sheet manufacturing apparatus **100** includes, for example, first connection pipes **140**, branch pipes **142**, first blowers **150**, second blowers **152**, and humidity sensors **160**.

(54) The humidifier **120** is, for example, an evaporative humidifier that causes moisture vaporization. The humidifier **120** humidifies a predetermined area **130** of the sheet manufacturing apparatus **100**. For example, a plurality of humidifiers **120** may be provided. In the illustrated example, three humidifiers **120a**, **120b**, and **120c** are provided as the humidifiers **120**. The humidifiers **120a**, **120b**, and **120c** are arranged, for example, in a predetermined direction.

(55) The predetermined area **130** humidified by the humidifiers **120** is, for example, a coarse crushing area **132** in which a raw material is coarsely crushed, a sorting area **134** in which the defibered substance having been coarsely crushed and defibered is sorted, a sheet forming area **136** in which the sheet S is formed by accumulating the defibered substance having been sorted, or a cutting area **138** in which the formed sheet S is cut.

(56) The coarse crushing area **132** includes the coarse crushing section **12** and a housing **133**. The housing **133** houses the coarse crushing section **12**. The sorting area **134** includes the sorting section **30**, the first web forming section **40**, and a housing **135**. The housing **135** houses the sorting section **30** and the first web forming section **40**. The sheet forming area **136** has the accumulation section **60**, the second web forming section **70**, the sheet forming section **80**, and a housing **137**. The housing **137** houses the accumulation section **60**, the second web forming section **70**, and the sheet forming section **80**. The cutting area **138** includes the cutting section **90** and a housing **139**. The housing **139** houses the cutting section **90**.

(57) The first connection pipes **140** couple the humidifiers **120** and the predetermined area **130** to each other. A plurality of first connection pipes **140** are provided. In the illustrated example, four first connection pipes **140** are provided as first connection pipes **140a**, **140b**, **140c**, and **140d**.

(58) The first connection pipe **140a** couples the humidifier **120a** and the coarse crushing area **132** to each other. The first connection pipe **140b** couples the humidifier **120a** and the cutting area **138** to each other. The first connection pipe **140c** couples the humidifier **120b** and the sheet forming area **136** to each other. The first connection pipe **140d** couples the humidifier **120c** and the sorting area **134** to each other. Furthermore, the humidifier **120c** is coupled to the pipe 5 by a connecting pipe **141**.

(59) The branch pipes **142** are branched from portions of the first connection pipes **140** between the first blowers **150** and the second blowers **152**. In other words, the branch pipes **142** are branched from portions of the first connection pipes **140** between the first blowers **150** and the second blowers **152**. The sheet manufacturing apparatus **100** can suck outside air through the branch pipes **142**.

(60) The first blowers **150** and the second blowers **152** are provided in the first connection pipes **140**. The first blowers **150** are located closer to the humidifier **120** than the second blowers **152** in the path of the first connection pipes **140**. In the illustrated example, the first blowers **150** are provided in the connection portions between the first connection pipes **140** and the humidifier **120**. The second blowers **152** are located closer to the predetermined area **130** than the first blowers **150** in the paths of the first connection pipes **140**. The humidifier **120** can humidify the predetermined area **130** by feeding moisture to the predetermined area **130** with the first blowers **150** and the second blowers **152** driven.

(61) The humidity sensors **160** are provided in the predetermined areas **130**. In the illustrated example, one humidity sensor **160** is provided in each of the housings **133**, **135**, and **139**, and two humidity sensors **160** are provided in the housing **137**. One of the two humidity sensors **160** provided in the housing **137** is provided in a portion close to the accumulation section **60**, and the other is provided in a portion close to the heating section **84**. The humidity sensor **160** detects the humidity of the predetermined area **130**. The humidity sensor **160** may further detect the temperature of the predetermined area **130**. It should be noted that the sheet forming area **136** may be divided into an area including the accumulation section **60** and the pressurizing section **82** and an area including the heating section **84**.

#### (62) 1.1.3 Control Section

(63) The sheet manufacturing apparatus **100** includes the control section **170**. The control section **170** includes a computer having, for example, a processor, a main storage device, and an input/output interface for inputting and outputting signals to and from the outside. The control section **170** performs various functions by, for example, causing the processor to execute a program read into the main storage device. Specifically, the control section **170** controls the manufacturing section **110** of the sheet manufacturing apparatus **100**. It should be noted that the control section **170** may include a combination of a plurality of circuits instead of the computer.

#### (64) 1.2 Control Method for Sheet Manufacturing Apparatus

(65) Next, a control method for the sheet manufacturing apparatus **100** according to the first embodiment will be described with reference to the drawings. FIG. **2** is a flowchart for describing the control method for the sheet manufacturing apparatus **100** according to the first embodiment.

(66) The user inputs a stop signal for stopping the manufacturing of the sheet **S** to the sheet manufacturing apparatus **100** that manufactures the sheet **S**. Specifically, the user inputs the stop signal for stopping the operation of the manufacturing section **110** to the control section **170** by operating an operation section, which is not illustrated. The stop signal includes, for example, the data of a desired stop time. The desired stop time is the time desired by the user from when the stop signal is input to the control section **170** until the operation of the manufacturing section **110** is stopped. The operation section includes, for example, a mouse, a keyboard, a touch panel, and the like. The control section **170** starts the processing when receiving the stop signal.

(67) First, in step **S10**, the control section **170** causes the supply section **10** to stop the supply of a raw material, as illustrated in FIG. **2**.

(68) Next, in step **S11**, in accordance with the detection values of the humidity sensors **160** provided in the predetermined areas **130**, the control section **170** determines whether the predetermined areas **130** need to be dried.

(69) Specifically, the control section **170** acquires the detection value of each of the humidity sensors **160** and determines that the predetermined area **130** does not need to be dried when the detection value acquired from the humidity sensor **160** is smaller than or equal to a predetermined value or determines that the predetermined area **130** needs to be dried when the detection value acquired from the humidity sensor **160** is larger than the predetermined value. When there are a plurality of predetermined areas **130**, the control section **170** determines whether each of the predetermined areas **130** needs to be dried.

(70) In step **S12**, when determining that the predetermined area **130** needs to be dried, that is, the result in step **S11** is YES, the control section **170** calculates the drying time required to dry the predetermined area **130** in accordance with the detection value of the humidity sensor **160**.

(71) Specifically, a storage section, which is not illustrated, stores a table illustrating the relationship between the detection value of the humidity sensor **160** and the drying time of the predetermined area **130**, and the control section **170** calculates the time required to dry the predetermined area **130** by verifying the acquired detection value of the humidity sensor **160** against this table.

(72) Next, in step **S13**, the control section **170** displays the calculated time in a display section,

which is not illustrated. The user is able to see the drying time by looking at the display section. The display section includes, for example, a liquid crystal display (LCD), an organic EL (electroluminescence) display, an EPD (electrophoretic display), a touch panel display, or the like.

(73) Next, in step **S14**, the control section **170** dries the predetermined area **130** with the outside air by controlling the first blowers **150** and the second blowers **152**.

(74) Specifically, the control section **170** blows the outside air supplied through the branch pipe **142** to the predetermined area **130** by driving the second blowers **152** with the driving of the first blowers **150** stopped as illustrated in FIG. 3. The predetermined area **130** is dried with the outside air. Before performing step **S14**, the first blowers **150** and the second blowers **152** may or may not be driven.

(75) It should be noted that FIG. 3 is a sectional view schematically illustrating the sheet manufacturing apparatus **100** and describes the control method for the sheet manufacturing apparatus **100**. In FIG. 3, the flows of the outside air supplied through the branch pipes **142** are indicated by arrows **A1**. In addition, in FIG. 3, the first blowers **150** that are not driven are indicated by circles with crosses.

(76) Next, in step **S15**, the control section **170** determines whether the drying of the predetermined area **130** is completed.

(77) Specifically, the control section **170** acquires the detection value of each of the humidity sensors **160** and determines that the drying of the predetermined area **130** is completed when the acquired detection value of the humidity sensor **160** is equal to or smaller than the predetermined value or determines that the drying of the predetermined area **130** is not completed when the acquired detection value of the humidity sensor **160** is larger than the predetermined value. The predetermined value in step **S15** may be the humidity of the outside air.

(78) In step **S16**, when determining that the drying of the predetermined area **130** is not completed, that is, the result in step **S15** is NO, the control section **170** adjusts the output of the second blowers **152** in accordance with the detection value of the humidity sensor **160** acquired in step **S15**.

(79) Specifically, the control section **170** determines whether the drying of the predetermined area **130** is completed within the time displayed in the display section in step **S13** by verifying the acquired detection value of the humidity sensor **160** against the table stored in the storage section. When determining that the drying of the predetermined area **130** is not completed within the time displayed in the display section, the control section **170** increases the output of the second blowers **152**. When determining that the drying of the predetermined area **130** is completed within the time displayed in the display section, the control section **170** maintains the output of the second blowers **152**. The control section **170** repeats steps **S15** and **S16** until it is determined in step **S15** that the drying of the predetermined area **130** is completed.

(80) In step **S17**, when determining that the drying of the predetermined area **130** is completed, that is, the result in step **S15** is YES, the control section **170** stops the driving of the second blowers **152**. When there are a plurality of predetermined areas **130**, if determining that the drying is completed in all of the predetermined areas **130**, the control section **170** stops the driving of the second blowers **152**.

(81) Next, in step **S18**, the control section **170** determines whether the desired stop time has elapsed after the stop signal is input.

(82) In step **S19**, when determining that the desired stop time has elapsed after the stop signal is entered, that is, the result in step **S18** is YES, the control section **170** stops the operation of the manufacturing section **110**. The control section **170** stops the operation of the manufacturing section **110** even when, for example, the raw material for manufacturing the sheet S remains in the manufacturing section **110**. Then, the control section **170** ends the processing.

(83) In step **S20**, when determining that the desired stop time has not elapsed after the stop signal is input, that is, the result in step **S18** is NO, the control section **170** determines whether the raw material remains in the manufacturing section **110**. For example, the manufacturing section **110** has

a sensor, not illustrated, that detects the raw material, and the control section **170** determines whether the raw material remains in the manufacturing section **110** in accordance with the detection result of the sensor.

(84) When determining that the raw material remains in the manufacturing section **110**, that is, the result in step **S20** is YES, the control section **170** returns the processing to step **S18**. The control section **170** repeats steps **S18** and **S20** until it is determined in step **S18** that the desired stop time has been elapsed.

(85) When determining that the raw material does not remain in the manufacturing section **110**, that is, the result in step **S20** is NO, the control section **170** proceeds to step **S19**. Then, the control section **170** ends the processing.

(86) When determining that drying of the predetermined area **130** is not necessary, that is, the result in step **S11** is NO, the control section **170** proceeds to step **S18**. Then, the control section **170** performs step **S19** and ends the processing.

(87) 1.3 Operation and Effect

(88) The control method for the sheet manufacturing apparatus **100** includes a stop signal input step of inputting, to the sheet manufacturing apparatus **100** manufacturing the sheet S, the stop signal stopping manufacturing of the sheet and a drying step of drying the predetermined area **130** of the sheet manufacturing apparatus **100** by controlling the blowers **150** and **152** of the sheet manufacturing apparatus **100** after the stop signal input step. Accordingly, in the control method for the sheet manufacturing apparatus **100**, the humidity within the sheet manufacturing apparatus **100** can be reduced after the manufacturing of the sheet S is stopped. This can reduce the possibility that rust forms or biological contaminants proliferate due to moisture after the manufacturing of sheet S is stopped.

(89) The control method for the sheet manufacturing apparatus **100** further includes a determination step of determining whether the predetermined area **130** needs to be dried in accordance with the detection value of the humidity sensor **160** provided in the predetermined area **130**, in which the drying step is performed when it is determined in the determination step that the predetermined area **130** needs to be dried. Accordingly, since the drying process is performed only when the predetermined area **130** needs to be dried in the control method for the sheet manufacturing apparatus **100**, power can be further saved compared with the case in which the drying step is always performed without the determination step. It should be noted that the control method may have a sequence such that the dry step is always performed when the stop signal is input without including the determination step of determining whether the predetermined area **130** needs to be dried.

(90) The control method for the sheet manufacturing apparatus **100** includes the step of calculating the time required for the drying step in accordance with the detection value of the humidity sensor **160**. Accordingly, in the control method for the sheet manufacturing apparatus **100**, the user can be informed of the time required for the drying step.

(91) In the control method for the sheet manufacturing apparatus **100**, the sheet manufacturing apparatus **100** includes the humidifiers **120** that humidify the predetermined areas **130** and the first connection pipes **140** that couple the humidifiers **120** and the predetermined area **130** to each other. Furthermore, the sheet manufacturing apparatus **100** includes the first blowers **150** provided in the first connection pipes **140**, the second blowers **152** provided in the first connection pipes **140** and located closer to the predetermined areas **130** than the first blowers **150** in the paths of the first connection pipes **140**, and the branch pipes **142** branched from portions of the first connection pipes **140** between the first blowers **150** and the second blowers **152**. Accordingly, in the control method for the sheet manufacturing apparatus **100**, the predetermined areas **130** can be humidified by the humidifiers **120**. This can prevent, for example, the fibers from adhering to the sheet manufacturing apparatus **100** due to static electricity.

(92) In the control method for the sheet manufacturing apparatus **100**, in the drying step, the

outside air supplied through the branch pipe **142** is blown to the predetermined area **130** by the second blowers **152** being driven with the driving of the first blower **150** stopped. Accordingly, in the control method for the sheet manufacturing apparatus **100**, the predetermined area **130** can be dried with the outside air.

(93) In the control method for the sheet manufacturing apparatus **100**, each of the predetermined areas **130** is the coarse crushing area **132** in which the raw material is coarsely crushed, the sorting area **134** in which the defibered substance having been coarsely crushed and defibered is sorted, the sheet forming area **136** in which the sheet S is formed by accumulating the defibered substance having been sorted, or the cutting area **138** in which the formed sheet S is cut. Accordingly, in the control method for the sheet manufacturing apparatus **100**, at least one of the coarse crushing area **132**, the sorting area **134**, the sheet forming area **136**, and the cutting area **138** can be dried.

## 2. Second Embodiment

### (94) 2.1 Sheet Manufacturing Apparatus

(95) Next, a sheet manufacturing apparatus according to a second embodiment will be described with reference to the drawings. FIG. **4** is a diagram schematically illustrating a sheet manufacturing apparatus **200** according to the second embodiment.

(96) The structural members of the sheet manufacturing apparatus **200** according to the second embodiment that have the same functions as the structural members of the sheet manufacturing apparatus **100** according to the first embodiment are denoted by the same reference numerals and the detailed descriptions thereof are omitted.

(97) The sheet manufacturing apparatus **200** is different from the sheet manufacturing apparatus **100** in that the manufacturing section **110** includes a second connection pipe **144** as illustrated in FIG. **4**.

(98) The second connection pipe **144** couples the suction mechanism **43** and the humidifier **120a** to each other. The suction mechanism **43** is a suction section that sucks the heat generated in the defibering section **20**. The defibering section **20** generates heat when defibering the raw material. The defibering section **20** is, for example, the hottest portion in the sheet manufacturing apparatus **100**.

### (99) 2.2 Control Method for Sheet Manufacturing Apparatus

(100) Next, a control method for the sheet manufacturing apparatus **200** according to the second embodiment will be described with reference to the drawings. FIG. **5** is a flowchart for describing the control method for the sheet manufacturing apparatus **200** according to the second embodiment.

(101) The differences between the control method for the sheet manufacturing apparatus **200** according to the second embodiment and the control method for the sheet manufacturing apparatus **100** according to the first embodiment will be described below and the similarities therebetween will be briefly described or not described.

(102) The processing for stopping the supply of the raw material in step **S30**, the processing for determining whether the predetermined areas **130** need to be dried in step **S31**, the processing for calculating the drying time in step **S32**, and the processing for displaying the drying time in step **S33**, which are illustrated in FIG. **5**, are basically the same as steps **S10** to **S13** described above.

(103) Next, in step **S34**, the control section **170** discharges the moisture in the humidifier **120** through the branch pipe **142** by driving the first blowers **150** with the driving of the second blowers **152** stopped as illustrated in FIG. **6**. Before performing step **S34**, the first blower **150** and the second blowers **152** may or may not be driven. The heat of the defibering section **20** sucked by the suction mechanism **43** in step **S34** is transmitted to the humidifier **120a**. For example, the humidifier **120b** communicates with the humidifier **120a**, and the humidifier **120c** communicates with the humidifier **120b**. Accordingly, the heat of the defibering section **20** sucked by the suction mechanism **43** can be transmitted to the humidifiers **120a**, **120b**, and **120c**.

(104) It should be noted that FIG. **6** is a sectional view schematically illustrating the sheet manufacturing apparatus **200** and describing the control method for the sheet manufacturing

apparatus **200**. In FIG. **6**, the flows of moisture in the humidifiers **120** is indicated by arrows **A2**. In addition, in FIG. **6**, the second blowers **152** that are not driven are illustrated by circles with crosses.

(105) Next, in step **S35**, the control section **170** dries the predetermined area **130** by controlling the first blowers **150** and the second blowers **152**.

(106) Specifically, as illustrated in FIG. **7**, the control section **170** transmits the heat of the defibering section **20** sucked by the suction mechanism **43** to the predetermined area **130** by driving the first blowers **150** and the second blowers **152**. The predetermined area **130** is dried with this heat.

(107) It should be noted that FIG. **7** is a diagram schematically illustrating the sheet manufacturing apparatus **200** and describing the control method for the sheet manufacturing apparatus **200**. In FIG. **7**, the heat of the defibering section **20** sucked by the suction mechanism **43** is indicated by arrow **A3**.

(108) Next, in step **S36**, the control section **170** determines whether the drying of the predetermined area **130** is completed. Step **S36** is basically the same as step **S15** described above.

(109) In step **S37**, when determining that the drying of the predetermined area **130** is not completed, that is, the result in step **S36** is NO, the control section **170** acquires the detection values of the humidity sensors **160** and adjusts the outputs of the first blowers **150** and the second blowers **152** in accordance with the acquired detection values of the humidity sensor **160**. The control section **170** repeats steps **S36** and **S37** until it is determined in step **S36** that the drying of the predetermined area **130** is completed.

(110) In step **S38**, when determining that the drying of the predetermined area **130** is completed, that is, the result in step **S36** is YES, the control section **170**, the control section **170** stops the driving of the first blower **150** and the second blowers **152**.

(111) The processing for determining whether the desired stop time has been elapsed in step **S39**, the processing for stopping the operation of the manufacturing section **110** in step **S40**, and the processing for determining whether the raw material of the sheet **S** remains in the manufacturing section **110** in step **S41** that follow are basically the same as steps **S18** to **S20** described above.

(112) 2.3 Operation and Effect

(113) In the control method for the sheet manufacturing apparatus **200**, the sheet manufacturing apparatus **200** includes the defibering section **20** that defibers the raw material, the suction mechanism **43** that sucks the heat generated in the defibering section **20**, and the second connection pipe **144** that couples the suction mechanism **43** and the humidifiers **120** to each other, and the humidifiers **120** are evaporative humidifiers that cause moisture vaporization. Accordingly, in the control method for the sheet manufacturing apparatus **200**, the heat generated in the defibering section **20** can be transmitted to the humidifiers **120**. This can compensate for the heat taken away by vaporizing moisture in the humidifiers **120**.

(114) The control method for the sheet manufacturing apparatus **200** includes a step of discharging the moisture in the humidifiers **120** through the branch pipes **142** by driving the first blowers **150** with the driving of the second blowers **152** stopped before the drying step and, in the drying step, the heat sucked by the suction mechanism **43** is transmitted to the predetermined area **130** by the first blowers **150** and the second blowers **152** being driven. Accordingly, in the control method for the sheet manufacturing apparatus **200**, the predetermined areas **130** can be dried with the heat generated in the defibering section **20**.

### 3. Third Embodiment

#### (115) 3.1 Sheet Manufacturing Apparatus

(116) Next, a sheet manufacturing apparatus according to a third embodiment will be described with reference to the drawings. FIG. **8** is a diagram schematically illustrating a sheet manufacturing apparatus **300** according to the third embodiment.

(117) The structural members of the sheet manufacturing apparatus **300** according to the third

embodiment that have the same functions as the structural members of the sheet manufacturing apparatus **100** according to the first embodiment and the sheet manufacturing apparatus **200** according to the second embodiment are denoted by the same reference numerals and the detailed descriptions thereof are omitted.

(118) The sheet manufacturing apparatus **300** is different from the sheet manufacturing apparatus **200** in that the manufacturing section **110** includes an environmental humidity sensor **162** as illustrated in FIG. **8**.

(119) The environmental humidity sensor **162** is provided outside the predetermined area **130**. The environmental humidity sensor **162** detects the environmental humidity outside the predetermined areas **130**. The environmental humidity sensor **162** may also be able to detect the environmental temperature.

(120) 3.2 Control Method for the Sheet Manufacturing Apparatus

(121) Next, the sheet manufacturing apparatus **300** according to the third embodiment will be described with reference to the drawings. FIG. **9** is a flowchart for describing the control method for the sheet manufacturing apparatus **300** according to the third embodiment.

(122) The differences between the control method for the sheet manufacturing apparatus **300** according to the third embodiment and the control method for the sheet manufacturing apparatus **100** according to the first embodiment and the differences between the control method for the sheet manufacturing apparatus **300** according to the third embodiment and the control method for the sheet manufacturing apparatus **200** according to the second embodiment will be described below and the similarities thereof are briefly described or are not described.

(123) The processing for stopping the supply of the raw material in step **S50** and the processing for determining whether the predetermined areas **130** need to be dried in step **S51**, which are illustrated in FIG. **9**, are basically the same as steps **S10** and **S11** described above.

(124) Next, in step **S55**, the control section **170** determines whether the detection value of the humidity sensor **160** provided in the sorting area **134** is larger than the detection value of the environmental humidity sensor **162**, which detects the environmental humidity, as illustrated in FIG. **9**.

(125) Specifically, the control section **170** acquires the detection value of the humidity sensor **160** provided in the sorting area **134** and the detection value of the environmental humidity sensor **162** and determines whether the acquired detection value of the humidity sensor **160** is larger than the acquired detection value of the environmental humidity sensor **162**.

(126) In step **S53**, when determining that the detection value of the humidity sensor **160** is larger than the detection value of the environmental humidity sensor **162**, that is, the result in step **S52** is YES, the control section **170** performs drying in pattern **1**. Specifically, the control section **170** performs the processing of steps **S12** to **S17** illustrated in FIG. **2**. Accordingly, the predetermined area **130** can be dried with the outside air.

(127) In step **S53**, when determining that the detection value of the humidity sensor **160** is not larger than the detection value of the environmental humidity sensor **162**, that is, the result in step **S52** is NO, the control section **170** performs drying in pattern **2**. Specifically, the control section **170** performs the processing of steps **S32** to **S38** illustrated in FIG. **5**. This can dry the predetermined area **130** with the heat generated in the defibering section **20**.

(128) The processing for determining whether the desired stop time has been elapsed in step **S55**, the processing for stopping the operation of the manufacturing section **110** in step **S56**, and the processing for determining whether the raw material of the sheet **S** remains in the manufacturing section **110** in step **S57** that follow are basically the same as steps **S18** to **S20** described above.

(129) 3.3 Operation and Effect

(130) The control method for the sheet manufacturing apparatus **300** includes the humidity determination step of determining whether the detection value of the humidity sensor **160** is larger than the detection value of the environmental humidity sensor **162**, which detects the

environmental humidity, after the determination step and before the drying step. When it is determined that the detection value of the humidity sensor **160** is larger than the detection value of the environmental humidity sensor **162** in the humidity determination step, the predetermined area **130** is dried with the outside air in the drying step. When it is determined that the detection value of the humidity sensor **160** is not larger than the detection value of the environmental humidity sensor **162** in the humidity determination step, the predetermined area **130** is dried with the heat generated in the defibering section **20** in the drying step. Accordingly, the control step for the sheet manufacturing apparatus **300** can efficiently dry the predetermined area.

#### 4. Fourth Embodiment

##### (131) 4.1 Sheet Manufacturing Apparatus

(132) Next, a sheet manufacturing apparatus **300** according to a fourth embodiment will be described with reference to the drawings. FIG. **10** is a diagram schematically illustrating a sheet manufacturing apparatus **400** according to the fourth embodiment.

(133) The structural members of the sheet manufacturing apparatus **400** according to the fourth embodiment that have the same functions as the structural members of the sheet manufacturing apparatus **100** according to the first embodiment and the sheet manufacturing apparatus **200** according to the second embodiment are denoted by the same reference numerals and the detailed descriptions thereof are omitted.

(134) The sheet manufacturing apparatus **400** is different from the sheet manufacturing apparatus **200** in that the manufacturing section **110** has heaters **180** as illustrated in FIG. **10**.

(135) As illustrated in FIG. **10**, the heaters **180** are provided in the humidifiers **120**. The heater **180** may be provided between a suction section and a filter, which are not illustrated, of the humidifier **120**. The heater **180** is not particularly limited, but may be, for example, a ceramic heater, a carbon heater, a halogen heater, a sheathed heater, or the like.

##### (136) 4.2 Control Method for Sheet Manufacturing Apparatus

(137) Next, the control method for the sheet manufacturing apparatus **400** according to the fourth embodiment will be described. The control method for the sheet manufacturing apparatus **400** according to the fourth embodiment is basically the same as the control method for the sheet manufacturing apparatus **200** according to the second embodiment illustrated in FIG. **5** except that the heaters **180** are driven at the same time with the driving of the first blowers **150** in step S34 and the driving of the heaters **180** is stopped at the same time with the stopping of the first blowers **150** and the second blowers **152** in step S38 in the control method for the sheet manufacturing apparatus **200**. Accordingly, the detailed description thereof are omitted.

##### (138) 4.3 Effect and Operation

(139) In the sheet manufacturing apparatus **400**, the manufacturing section **110** includes the humidifiers **120** that humidify the predetermined areas **130** and the heaters **180** provided in the humidifiers **120**. Accordingly, the sheet manufacturing apparatus **400** can efficiently dry a wet portion in the humidifier **120** by the heat of the heater **180** when moisture in the humidifier **120** is discharged through the branch pipe **142**. Furthermore, by transmitting the heat of the heater **180** to the predetermined area **130** when the heat generated in the defibering section **20** is transmitted to the predetermined area **130**, even when condensation occurs in the predetermined area **130**, the portion wet by the condensation can be efficiently dried.

(140) The embodiments and modifications described above are only examples, and the present disclosure is not limited thereto. For example, the embodiments and the modifications can be combined with each other as appropriate.

(141) The present disclosure includes a structure substantially the same as the structures described in the embodiments. That is, the present disclosure includes, for example, a structure having the same function, method, and result or a structure having the same object and effect. In addition, the present disclosure also includes a structure in which non-essential parts of the structures described in the embodiments have been replaced. In addition, the present disclosure includes a structure

having the same operation and effect as the structures described in the embodiments or a structure capable of achieving the same object. In addition, the present disclosure includes a structure obtained by adding a known technique to the structures described in the embodiments.

(142) What is derived from the embodiments and modifications described above will be described below.

(143) The control method for a sheet manufacturing apparatus according to an aspect of the present disclosure includes: the stop signal input step of inputting, to the sheet manufacturing apparatus manufacturing the sheet, the stop signal stopping manufacturing of the sheet; and the drying step of drying the predetermined area of the sheet manufacturing apparatus by controlling the blower of the sheet manufacturing apparatus after the stop signal input step.

(144) This control method for the sheet manufacturing apparatus can reduce the possibility that rust forms or biological contaminants proliferate due to moisture after the manufacturing of the sheet is stopped.

(145) The control method for a sheet manufacturing apparatus according to an aspect may further include the determination step of determining whether the predetermined area needs to be dried in accordance with the detection value of the humidity sensor provided in the predetermined area, in which the drying step may be performed when it is determined in the determination step that the predetermined area needs to be dried.

(146) According to this control method for the sheet manufacturing apparatus, since the drying step is performed only when the predetermined area needs to be dried, power can be saved compared with the case in which the drying step is always performed without the determination step.

(147) The control method for a sheet manufacturing apparatus according to an aspect may further include calculating a time required for the drying step in accordance with the detection value of the humidity sensor.

(148) According to this control method for the sheet manufacturing apparatus, it is possible to inform the user of the time required for the drying step.

(149) In the control method for a sheet manufacturing apparatus, the sheet manufacturing apparatus may include the humidifier humidifying the predetermined area, the first connection pipe coupling the humidifier and the predetermined area to each other, the first blower as the blower, the first blower being provided in the first connection pipe, the second blower as the blower, the second blower being provided in the first connection pipe, the second blower being located closer to the predetermined area than the first blower in the path of the first connection pipe, and the branch pipe branched from the first connection pipe between the first blower and the second blower.

(150) According to this control method for the sheet manufacturing apparatus, the predetermined area can be humidified with the humidifier.

(151) In the control method for a sheet manufacturing apparatus according to an aspect, in the drying step, outside air supplied through the branch pipe may be blown to the predetermined area by the second blower being driven with the driving of the first blower stopped.

(152) This control method for the sheet manufacturing apparatus can dry the predetermined area with the outside air.

(153) In the control method for a sheet manufacturing apparatus according to an aspect, the sheet manufacturing apparatus may include the defibering section defibering the raw material, the suction section sucking heat generated in the defibering section, the second connection pipe coupling the suction section and the humidifier to each other, in which the humidifier may be the evaporative humidifier that causes moisture vaporization.

(154) According to this control method for the sheet manufacturing apparatus, the heat taken by vaporizing moisture in the humidifier can be compensated for.

(155) The control method for a sheet manufacturing apparatus according to an aspect may further include discharging moisture in the humidifier through the branch pipe by driving the first blower with driving of the second blower stopped before the drying step, in which the drying step may

transmit the heat sucked by the suction section to the predetermined area by driving the first blower and the second blower.

(156) According to this control method for the sheet manufacturing apparatus, the predetermined area can be dried with the heat generated in the defibering section.

(157) The control method for a sheet manufacturing apparatus according to an aspect may further include the humidity determination step of determining whether the detection value of the humidity sensor is larger than the detection value of the environmental humidity sensor detecting environmental humidity after the determination step and before the drying step, in which the predetermined area may be dried with outside air when it is determined in the humidity determination step that the detection value of the humidity sensor is larger than the detection value of the environmental humidity sensor, and the predetermined area may be dried with the heat in the drying step when it is determined in the humidity determination step that the detection value of the humidity sensor is not larger than the detection value of the environmental humidity sensor.

(158) According to this control method for the sheet manufacturing apparatus, the predetermined area can be efficiently dried.

(159) In the control method for a sheet manufacturing apparatus according to an aspect, the predetermined area may be the coarse crushing area in which the raw material is coarsely crushed, the sorting area in which the defibered substance coarsely crushed and defibered is sorted, the sheet forming area in which the sheet is formed by accumulating the sorted defibered substance, or the cutting area in which the sheet is cut.

(160) According to this control method for the sheet manufacturing apparatus, at least one of the coarse crushing area, the sorting area, the sheet forming area, and the cutting area can be dried.

(161) The sheet manufacturing apparatus according to an aspect includes: the manufacturing section manufacturing the sheet; and the control section controlling the manufacturing section, in which the control section dries the predetermined area of the manufacturing section by controlling the blower of the manufacturing section when receiving the stop signal stopping operation of the manufacturing section.

(162) According to this control method for the sheet manufacturing apparatus, it is possible to reduce the possibility that rust forms or biological contaminants proliferate due to moisture after the manufacturing of sheets is stopped.

(163) In the sheet manufacturing apparatus according to an aspect, the manufacturing section may include the humidifier humidifying the predetermined area and the heater provided in the humidifier.

(164) According to this control method for the sheet manufacturing apparatus, the wet portion in the humidifier can be efficiently dried by the heat of the heater.

## Claims

1. A control method for a sheet manufacturing apparatus, the method comprising: a stop signal input step of inputting, to the sheet manufacturing apparatus manufacturing a sheet, a stop signal stopping manufacturing of the sheet; a drying step of drying a predetermined area of the sheet manufacturing apparatus by controlling a blower of the sheet manufacturing apparatus after the stop signal input step; and a determination step of determining whether the predetermined area needs to be dried in accordance with a detection value of a humidity sensor provided in the predetermined area, wherein the drying step is performed when it is determined in the determination step that the predetermined area needs to be dried.

2. The control method for a sheet manufacturing apparatus according to claim 1, further comprising: calculating a time required for the drying step in accordance with the detection value of the humidity sensor.

3. The control method for a sheet manufacturing apparatus according to claim 1, wherein the sheet

manufacturing apparatus includes a humidifier humidifying the predetermined area, a first connection pipe coupling the humidifier and the predetermined area to each other, a first blower as the blower, the first blower being provided in the first connection pipe, a second blower as the blower, the second blower being provided in the first connection pipe, the second blower being located closer to the predetermined area than the first blower in a path of the first connection pipe, and a branch pipe branched from the first connection pipe between the first blower and the second blower.

4. The control method for a sheet manufacturing apparatus according to claim 3, wherein in the drying step, outside air supplied through the branch pipe is blown to the predetermined area by the second blower being driven with the driving of the first blower stopped.

5. The control method for a sheet manufacturing apparatus according to claim 3, wherein the sheet manufacturing apparatus includes a defibering section for defibering a raw material, a suction section for sucking heat generated in a defibering section, and a second connection pipe coupling the suction section and the humidifier to each other, wherein the humidifier is an evaporative humidifier that causes moisture vaporization.

6. The control method for a sheet manufacturing apparatus according to claim 5, further comprising: discharging moisture in the humidifier through the branch pipe by driving the first blower with driving of the second blower stopped before the drying step, wherein in the drying step, the heat sucked by the suction section is transmitted to the predetermined area by the first blower and the second blower being driven.

7. The control method for a sheet manufacturing apparatus according to claim 5, further comprising: a humidity determination step of determining whether the detection value of the humidity sensor is larger than a detection value of an environmental humidity sensor detecting environmental humidity after the determination step and before the drying step, wherein the predetermined area is dried with outside air when it is determined in the humidity determination step that the detection value of the humidity sensor is larger than the detection value of the environmental humidity sensor, and the predetermined area is dried with the heat in the drying step when it is determined in the humidity determination step that the detection value of the humidity sensor is not larger than the detection value of the environmental humidity sensor.

8. The control method for a sheet manufacturing apparatus according to claim 1, wherein the predetermined area is a coarse crushing area in which a raw material is coarsely crushed, a sorting area in which a defibered substance coarsely crushed and defibered is sorted, a sheet forming area in which the sheet is formed by accumulating the sorted defibered substance, or a cutting area in which the sheet is cut.

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